

Nathaniel Oliver

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Education

M.S. Data Science, UC San Diego - Expected June 2027 | GPA: 4.0

B.S. Mathematics & Economics, Minor in Data Science, UC San Diego - June 2025

Experience

Data Science Intern, IDX Exchange [\[live demo\]](#)

January 2026 – Present

- **Built an end-to-end ML pipeline** to predict single-family home sale prices across California using on and off-market MLS data (100K+ observations), performing geospatial filtering, missingness mechanism analysis, and custom imputation strategies.
- **Engineered polynomial/interaction features, applied target encoding, and implemented geo-clustering** to capture non-linear, high-cardinality, and spatial pricing patterns.
- **Developed a 2-level stacked ensemble (XGBoost + LightGBM)** with GBRegressor meta-learner using segment-aware weighting and RandomizedSearchCV.
- **Achieved 7.74% MdAPE**, comparable to Redfin's model, observing minimal temporal drift.

Economic Research Assistant, UCSD Department of Economics

April 2024 – June 2025

- **Built reproducible ETL pipelines** to tidy, aggregate, and explore large, multi-source datasets (UNHCR, USCIS) using Python, R, and Stata, and documented them.
- **Used Regression and DiD Analysis (QGIS, Stata)** to segment farming populations in Africa and determine if refugee presence increased crop prices.
- **Utilized Time-Series and Breakpoint Analysis (R, Python)** to identify statistically significant structural changes in cohabitation behavior in the US from 2007 to 2024 across major metropolitan areas, and used regression analysis to determine the most significant demographic factors driving these changes.

Projects

Agentic Resume Tailoring System

2026 - Present

- **Built a multi-agent LangGraph pipeline** that ingests resumes, transcripts, LinkedIn profiles, GitHub metadata, and repos, and creates a knowledge graph, enabling indirect skill inference for ATS tailoring.
- **Implemented multi-strategy skill matching** using direct lookup, fuzzy name, graph traversal, and semantic similarity, with a reflection loop to prevent skill fabrication and ensure explainable ATS scores.
- **Delivered a CLI and a TUI** with a chat-based interface for seamless Resume Tailoring.

E-Commerce Browsing Session Revenue Prediction and Behavioral Segmentation [\[GitHub\]](#)

March 2026

- **Analyzed 12,000+ browsing sessions** using hypothesis testing (Mann-Whitney U, Chi-Squared) and feature importance from RF and XGBoost models to identify significant revenue drivers.
- **Built optimized Random Forest and XGBoost classifiers** on imbalanced data, achieving 90% accuracy and 80% recall on identifying revenue-generating sessions, while isolating strong behavioral predictors by ablating site-level metrics.
- **Applied K-Means clustering and PCA projection (Unsupervised Learning)** to identify five distinct session archetypes, revealing that progressive engagement depth characterizes the pathway to purchase.

Skills

Languages: Python, R, SQL, Spark

ML/AI: scikit-learn, PyTorch, TensorFlow, LangChain, LangGraph, XGBoost, LightGBM

Data & Vis: pandas, NumPy, SciPy, QGIS, Matplotlib, Seaborn

Databases & Cloud: DuckDB, PostgreSQL, MongoDB, AWS S3

Workflow: Git, Docker, Streamlit